CMS Open Data H to tau tau Search: Phase 4c Audit-Corrected Observed Results

Analysis my_analysis

2026-06-02

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Change Log

Phase 4c v2 audit correction

- Added a same-sign data-driven QCD/fake background estimate.
- Rebuilt the visible-mass baseline and retained the single D_NN classifier-score result.
- Removed historical optimized-score, add-MET, and duplicate DNN-alias branches from the current output set.

1 Phase 4c Observed Inference: Full Data

1.1 Summary

The full-data result is rebuilt from the current local data and MC inputs using two retained workspaces: the cut-based visible-mass baseline and the three-category $D_N N$ classifier-score workspace. Historical optimized-score, add-MET, and duplicate DNN aliases are not part of this output set.

1.2 W and QCD Control Inputs

The full-data W+jets scale from the high-mT control region is `0.9190 ± 0.0139`. The VBF background control scale from the VBF-like top-btag non-signal region is `0.3448 ± 0.0612`; it is applied only to MC backgrounds in the VBF fit category. This addresses the large VBF data/MC overprediction seen in the original observed templates without changing the signal normalization or the boosted/zero-jet categories. The same-sign QCD/fake estimate uses data minus non-QCD MC in the same-sign low-mT region and a transfer factor measured in the lowest fitted-observable bin. For the visible-mass baseline this factor is `0.6163 ± 0.1944`.

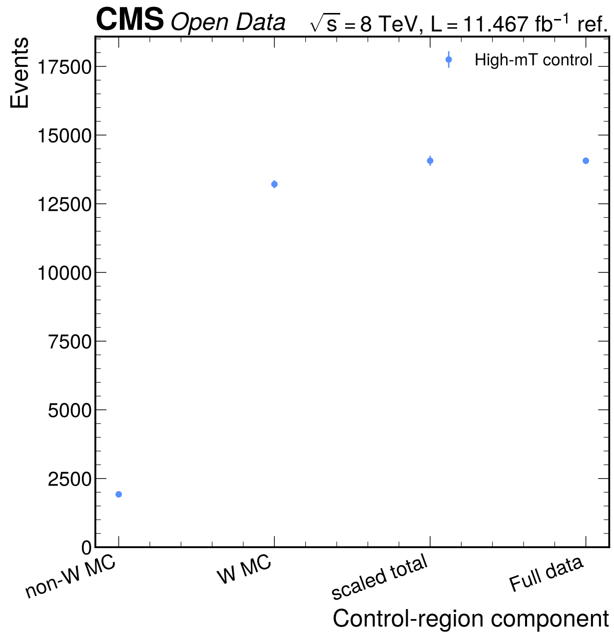

Figure 1: Full high-mT W control comparison. The figure shows non-W MC, nominal W MC, the scaled control-region prediction, and full data in the control region. The scale is derived outside the signal region and then propagated into the observed workspaces.

1.3 Visible-Mass Baseline

Category	Data	Background	QCD/fake	Data/background	Chi2/ndf	Max abs pull
vbf	70	53.37	14.70	1.312	1.999	2.01
boosted	1138	1249.00	28.15	0.911	0.492	1.07
zero_jet	25920	26732.68	1742.04	0.970	0.126	0.55

The visible-mass baseline fit gives `mu_hat = 2.4740`, an observed 95% CLs limit `mu < 5.9262`, and a diagnostic `q0` value `Z = 1.3259`. The median expected limit from the same workspace is `3.6919`.

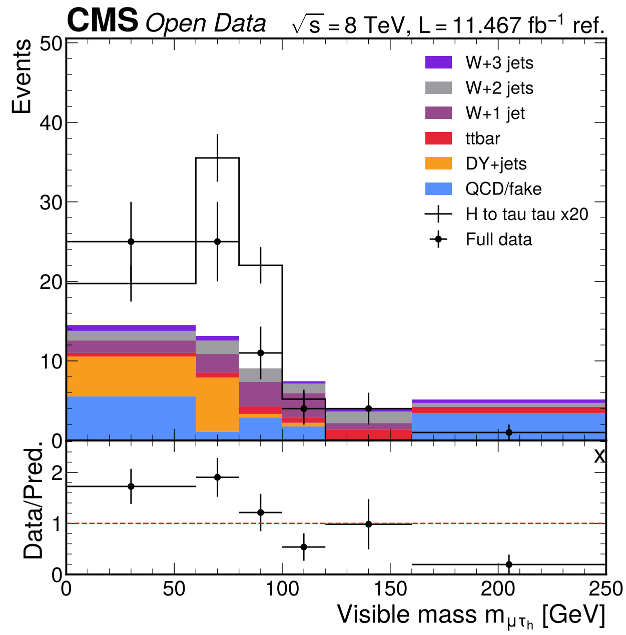

Figure 2: Visible-mass baseline validation in the VBF category. The plot compares localized Run2012B/C TauPlusX data to the visible-mass baseline model after the same-sign QCD/fake correction. This category remains statistically limited but is included in the simultaneous baseline fit.

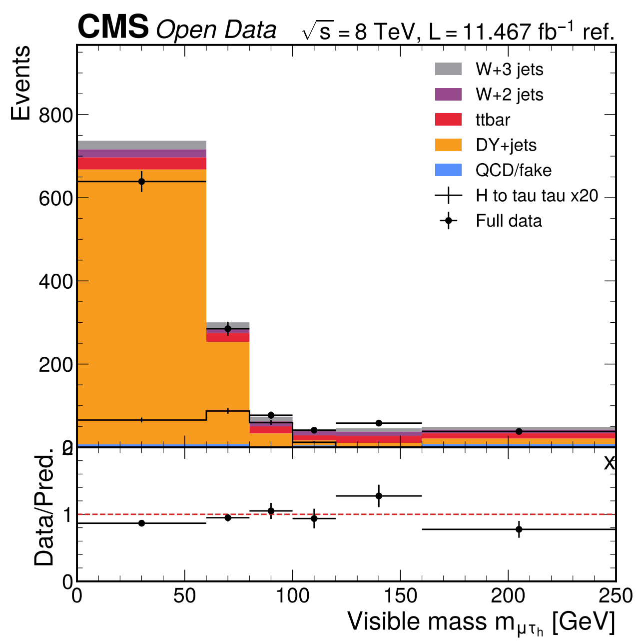

Figure 3: Visible-mass baseline validation in the boosted category. The plot compares full data to the visible-mass baseline model with the W high-mT scale and same-sign QCD/fake template. This category is used in the simultaneous baseline fit.

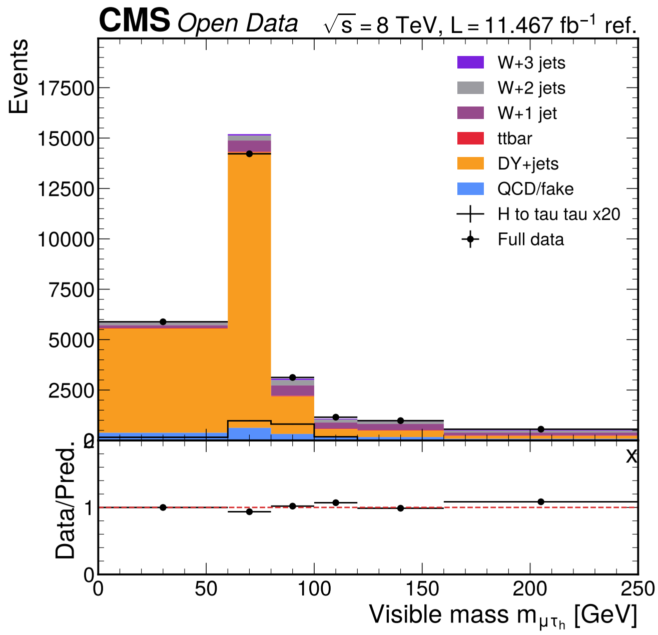

Figure 4: Visible-mass baseline validation in the zero-jet category. The zero-jet category contributes the largest data statistics to the visible-mass baseline fit and anchors the cut-based result.

1.4 D_NN Result

The retained NN result uses the XGBoost classifier score `mva_score_xgboost` in the same VBF, boosted, and zero-jet categories with 20 uniform score bins. It gives `mu_hat = 1.6160`, observed 95% CLs `mu < 3.5769`, and median expected limit `1.8069`.

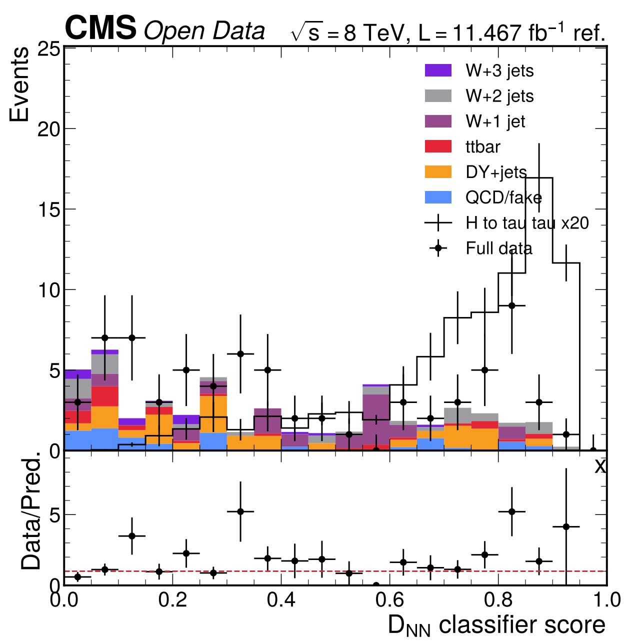

Figure 5: D_NN validation in the VBF category. The plot compares observed data to the full MC expectation for the retained classifier-score workspace. The same score definition and binning are used in all three categories.

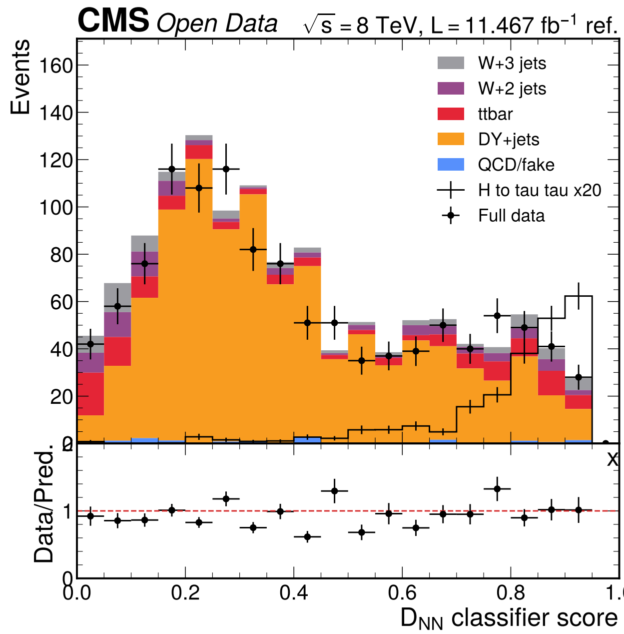

Figure 6: D_NN validation in the boosted category. The plot compares observed data to the retained classifier-score workspace in the boosted category. The lower panel reports the data-to-prediction ratio.

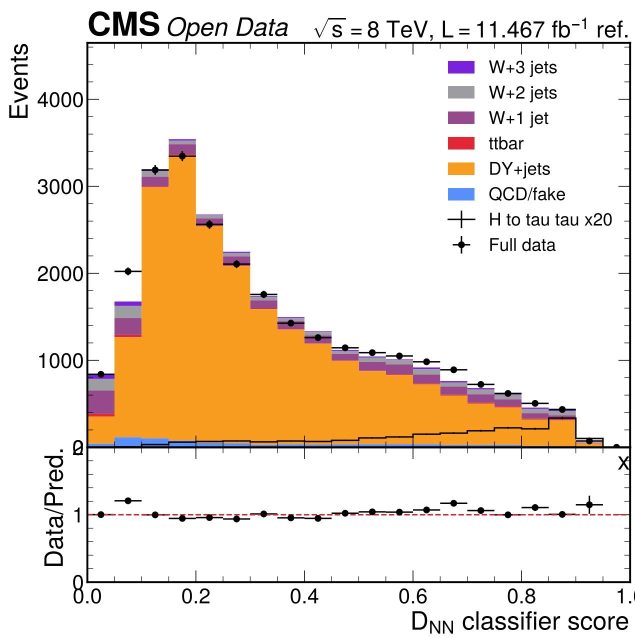

Figure 7: D_NN validation in the zero-jet category. The plot compares observed data to the retained classifier-score workspace in the statistically dominant zero-jet category.

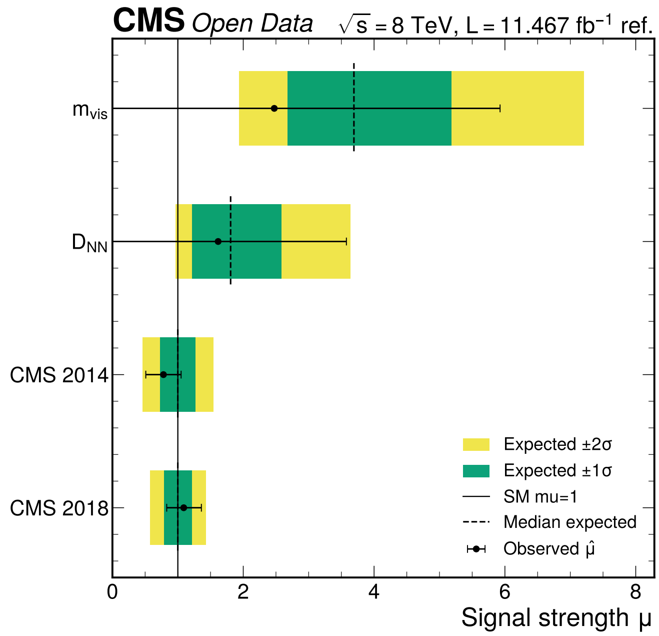

Figure 8: Observed limit and significance summary. The figure shows the primary visible-mass baseline and retained D_NN result on the same signal-strength scale.

References